

IT2212 - DISCRETE MATHEMATICS

Exam Preparation Worksheet

Topics Covered:

- Propositional Logic & Truth Tables
- Set Theory & Operations

Instructions:

- Complete all exercises on a separate sheet of paper
- Answer keys are provided after each section
- Show all work for full credit
- Time yourself - allow 2-3 hours for complete worksheet

PART 1: PROPOSITIONAL LOGIC & TRUTH TABLES

Section A: Identify Propositions (10 points)

Instructions: For each statement below, indicate whether it is a proposition (Y/N). If yes, state its truth value (T/F) or write 'indeterminate' if it cannot be determined.

1. The Philippines is an archipelago.
2. $x + 5 = 10$
3. Please close the door.
4. 15 is divisible by 3.
5. What time is it?
6. If $x > 5$, then $x > 3$.
7. Turn off your phone!
8. The square root of 16 is 4.
9. $y < 20$
10. All prime numbers are odd.

ANSWER KEY - Section A

Question	Proposition?	Truth Value
1	Y	T
2	N	--
3	N	--
4	Y	T
5	N	--
6	Y	T
7	N	--
8	Y	T
9	N	--
10	Y	F (2 is prime and even)

Section B: Complete Truth Tables (30 points)

Instructions: Complete the following truth tables by filling in all missing values.

B.1 Complete the truth table for $\sim P$ (3 points)

P	$\sim P$
T	
F	

B.2 Complete the truth table for $P \wedge Q$ (5 points)

P	Q	$P \wedge Q$
T	T	
T	F	
F	T	
F	F	

B.3 Complete the truth table for $P \vee Q$ (5 points)

P	Q	$P \vee Q$
T	T	
T	F	
F	T	
F	F	

B.4 Complete the truth table for $P \rightarrow Q$ (5 points)

P	Q	$P \rightarrow Q$
T	T	
T	F	
F	T	
F	F	

B.5 Complete the truth table for $P \leftrightarrow Q$ (5 points)

P	Q	$P \leftrightarrow Q$
T	T	
T	F	
F	T	
F	F	

B.6 Complete the truth table for $P \oplus Q$ (Exclusive OR) (7 points)

P	Q	$P \oplus Q$
T	T	
T	F	
F	T	
F	F	

ANSWER KEY - Section B

B.1 Answer:

P	$\sim P$
T	F
F	T

B.2 Answer:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

B.3 Answer:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

B.4 Answer:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

B.5 Answer:

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

B.6 Answer:

P	Q	$P \oplus Q$
T	T	F

T	F	T
F	T	T
F	F	F

Section C: Complex Truth Tables (25 points)

Instructions: Construct complete truth tables for the following compound propositions. Show all intermediate steps.

C.1 $\sim P \wedge Q$ (5 points)

P	Q	$\sim P$	$\sim P \wedge Q$
T	T		
T	F		
F	T		
F	F		

C.2 $(P \vee Q) \wedge \sim R$ (8 points)

P	Q	R	$P \vee Q$	$\sim R$	$(P \vee Q) \wedge \sim R$
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

C.3 $P \rightarrow (Q \vee R)$ (8 points)

P	Q	R	$Q \vee R$	$P \rightarrow (Q \vee R)$
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

ANSWER KEY - Section C

C.1 Answer:

P	Q	$\sim P$	$\sim P \wedge Q$
T	T	F	F
T	F	F	F
F	T	T	T
F	F	T	F

C.2 Answer:

P	Q	R	$P \vee Q$	$\sim R$	$(P \vee Q) \wedge \sim R$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	F	F
F	T	F	T	T	T
F	F	T	F	F	F
F	F	F	F	T	F

C.3 Answer:

P	Q	R	$Q \vee R$	$P \rightarrow (Q \vee R)$
T	T	T	T	T
T	T	F	T	T
T	F	T	T	T
T	F	F	F	F
F	T	T	T	T
F	T	F	T	T
F	F	T	T	T
F	F	F	F	T

Section D: Statement Analysis (15 points)

Instructions: Convert each statement into symbolic form using appropriate logical connectives.

1. If it is raining, then the ground is wet. (3 points)
2. You can pass the exam if and only if you study hard. (3 points)
3. Either you are 18 years old or you have parental consent. (3 points)
4. The number is not even and it is not odd. (3 points)
5. If you are a student, then you have an ID, and if you have an ID, then you can enter. (3 points)

ANSWER KEY - Section D

1. $P \rightarrow Q$

P = It is raining, Q = The ground is wet

2. $P \leftrightarrow Q$

P = You can pass the exam, Q = You study hard

3. $P \vee Q$

P = You are 18 years old, Q = You have parental consent

4. $\sim P \wedge \sim Q$

P = The number is even, Q = The number is odd (Note: This is a contradiction)

5. $(P \rightarrow Q) \wedge (Q \rightarrow R)$

P = You are a student, Q = You have an ID, R = You can enter

Section E: Logical Classification (20 points)

Instructions: For each proposition, construct a truth table and classify it as a Tautology, Contradiction, or Contingency.

E.1 $P \vee \sim P$ (Classify this proposition) (5 points)

P	$\sim P$	$P \vee \sim P$
T		
F		

Classification: _____

E.2 $P \wedge \sim P$ (Classify this proposition) (5 points)

P	$\sim P$	$P \wedge \sim P$
T		
F		

Classification: _____

E.3 $(P \rightarrow Q) \leftrightarrow (\sim P \vee Q)$ (Classify this proposition) (10 points)

P	Q	$P \rightarrow Q$	$\sim P$	$\sim P \vee Q$	$(P \rightarrow Q) \leftrightarrow (\sim P \vee Q)$
T	T				
T	F				
F	T				
F	F				

Classification: _____

ANSWER KEY - Section E

E.1 Answer:

P	$\sim P$	$P \vee \sim P$
T	F	T
F	T	T

Classification: TAUTOLOGY (All values are True)

E.2 Answer:

P	$\sim P$	$P \wedge \sim P$
T	F	F
F	T	F

Classification: CONTRADICTION (All values are False)

E.3 Answer:

P	Q	$P \rightarrow Q$	$\sim P$	$\sim P \vee Q$	$(P \rightarrow Q) \leftrightarrow (\sim P \vee Q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

Classification: TAUTOLOGY (All values are True)

This proves that $(P \rightarrow Q)$ is logically equivalent to $(\sim P \vee Q)$

PART 2: SET THEORY & OPERATIONS

Section F: Set Fundamentals (20 points)

Instructions: Answer the following questions about sets.

F.1 Let $A = \{2, 4, 6, 8, 10\}$. Is $5 \in A$? Is $6 \in A$? (2 points)

F.2 Write the set $B = \{x \mid x \text{ is a vowel}\}$ using roster method. (2 points)

F.3 Is $C = \{1, 2, 3\}$ equal to $D = \{3, 2, 1\}$? Explain. (2 points)

F.4 Let $E = \{a, b, c\}$. How many subsets does E have? (3 points)

F.5 Let $E = \{a, b, c\}$. How many proper subsets does E have? (3 points)

F.6 List all subsets of $F = \{x, y\}$. (4 points)

F.7 Is $\{1, 2\} \subseteq \{1, 2, 3, 4\}$? (2 points)

F.8 Is $\{1, 2\} \subset \{1, 2\}$? (2 points)

ANSWER KEY - Section F

F.1 $5 \notin A$ (5 is NOT an element of A); $6 \in A$ (6 IS an element of A)

F.2 $B = \{a, e, i, o, u\}$

F.3 Yes, $C = D$. Sets are equal if they contain exactly the same elements, regardless of order.

F.4 $2^3 = 8$ subsets

F.5 $2^3 - 1 = 8 - 1 = 7$ proper subsets

F.6 $\{\}, \{x\}, \{y\}, \{x, y\}$

F.7 Yes, $\{1, 2\} \subseteq \{1, 2, 3, 4\}$ (it is a subset)

F.8 No, $\{1, 2\} \not\subset \{1, 2\}$ (it is not a proper subset because they are equal)

Section G: Set Operations (30 points)

Instructions: Given the following sets, perform the indicated operations.

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{4, 5, 6, 7, 8\}$$

$$C = \{2, 4, 6, 8, 10\}$$

G.1 $A' =$ _____ (3 points)

G.2 $A \cap B =$ _____ (3 points)

G.3 $A \cup B =$ _____ (3 points)

G.4 $A - B =$ _____ (3 points)

G.5 $B - A =$ _____ (3 points)

G.6 $A \oplus B$ (Symmetric Difference) = _____ (4 points)

G.7 $(A \cup B)' =$ _____ (4 points)

G.8 $A \cap B \cap C =$ _____ (4 points)

G.9 $A \cup (B \cap C) =$ _____ (3 points)

ANSWER KEY - Section G

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{1, 2, 3, 4, 5\}$$

$$B = \{4, 5, 6, 7, 8\}$$

$$C = \{2, 4, 6, 8, 10\}$$

$$\mathbf{G.1} \ A' = \{6, 7, 8, 9, 10\}$$

Explanation: Elements in U but not in A

$$\mathbf{G.2} \ A \cap B = \{4, 5\}$$

Explanation: Elements in both A and B

$$\mathbf{G.3} \ A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

Explanation: All elements in A or B or both

$$\mathbf{G.4} \ A - B = \{1, 2, 3\}$$

Explanation: Elements in A but not in B

$$\mathbf{G.5} \ B - A = \{6, 7, 8\}$$

Explanation: Elements in B but not in A

$$\mathbf{G.6} \ A \oplus B = \{1, 2, 3, 6, 7, 8\}$$

Explanation: Elements in A or B but not in both

$$\mathbf{G.7} \ (A \cup B)' = \{9, 10\}$$

Explanation: Complement of the union

$$\mathbf{G.8} \ A \cap B \cap C = \{4\}$$

Explanation: Elements common to all three sets

$$\mathbf{G.9} \ A \cup (B \cap C) = \{1, 2, 3, 4, 5, 6, 8\}$$

Explanation: $B \cap C = \{4, 6, 8\}$, then union with A

Section H: Venn Diagram Problems (30 points)

Instructions: Use the Venn diagram information to answer the questions.

Problem H.1: A survey of 100 students found:

- 45 students study Mathematics (M)
- 38 students study Physics (P)
- 42 students study Chemistry (C)
- 15 students study both M and P
- 18 students study both M and C
- 12 students study both P and C
- 8 students study all three subjects

- a. How many students study only Mathematics? (5 points)
- b. How many students study only Physics? (5 points)
- c. How many students study only Chemistry? (5 points)
- d. How many students study exactly two subjects? (5 points)
- e. How many students study at least one of the three subjects? (5 points)
- f. How many students study none of the three subjects? (5 points)

HINT: Draw a Venn diagram with three overlapping circles. Start by filling in the center (all three), then work outward.

ANSWER KEY - Section H

Problem H.1 Solution:

Step-by-step solution:

1. Start with the center (all three subjects): 8 students
2. $M \cap P$ only: $15 - 8 = 7$ students
3. $M \cap C$ only: $18 - 8 = 10$ students
4. $P \cap C$ only: $12 - 8 = 4$ students
5. M only: $45 - (7 + 8 + 10) = 45 - 25 = 20$ students
6. P only: $38 - (7 + 8 + 4) = 38 - 19 = 19$ students
7. C only: $42 - (10 + 8 + 4) = 42 - 22 = 20$ students

a. 20 students study only Mathematics

b. 19 students study only Physics

c. 20 students study only Chemistry

d. $7 + 10 + 4 = 21$ students study exactly two subjects

e. $20 + 19 + 20 + 7 + 10 + 4 + 8 = 88$ students study at least one subject

f. $100 - 88 = 12$ students study none of the three subjects

Section I: Multiple Choice Questions (20 points)

Instructions: Select the best answer for each question.

I.1 Which of the following is a well-defined set? (2 points)

- A) The set of beautiful flowers
- B) The set of prime numbers less than 20
- C) The set of good students
- D) The set of large numbers

I.2 If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$, what is $A \cap B$? (2 points)

- A) $\{1, 2, 3, 4\}$
- B) $\{2, 3\}$
- C) $\{1, 4\}$
- D) $\{\}$

I.3 Which statement is a tautology? (2 points)

- A) $P \wedge Q$
- B) $P \vee \sim P$
- C) $P \rightarrow Q$
- D) $P \wedge \sim P$

I.4 If P is true and Q is false, what is the truth value of $P \rightarrow Q$? (2 points)

- A) True
- B) False
- C) Cannot be determined
- D) Both true and false

I.5 How many elements are in the power set of $\{a, b, c, d\}$? (2 points)

- A) 4
- B) 8
- C) 16
- D) 32

I.6 Which of the following represents the symmetric difference $A \oplus B$? (2 points)

- A) $(A \cup B) - (A \cap B)$
- B) $A \cap B$
- C) $A \cup B$
- D) $A - B$

I.7 If $A \subseteq B$ and $B \subseteq A$, then: (2 points)

- A) $A \subset B$
- B) $A = B$
- C) $A \cap B = \{\}$
- D) A and B are disjoint

I.8 The negation of 'All cats are mammals' is: (2 points)

- A) All cats are not mammals
- B) No cats are mammals
- C) Some cats are not mammals
- D) Some cats are mammals

I.9 Which logical connective has the highest precedence? (2 points)

- A) $\leftrightarrow$
- B) $\rightarrow$
- C) $\vee$
- D) $\sim$

I.10 If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2, 3\}$, what is A' ? (2 points)

- A) $\{1, 2, 3\}$
- B) $\{4, 5\}$
- C) $\{3, 4, 5\}$
- D) $\{\}$

ANSWER KEY - Section I

I.1 Answer: **B** - B) The set of prime numbers less than 20

I.2 Answer: **B** - B) {2, 3}

I.3 Answer: **B** - B) $P \vee \sim P$

I.4 Answer: **B** - B) False

I.5 Answer: **C** - C) 16

I.6 Answer: **A** - A) $(A \cup B) - (A \cap B)$

I.7 Answer: **B** - B) $A = B$

I.8 Answer: **C** - C) Some cats are not mammals

I.9 Answer: **D** - D) $\sim$

I.10 Answer: **B** - B) {4, 5}

Section J: Prove Logical Equivalence (15 points)

Instructions: Use truth tables to prove or disprove the following logical equivalences.

J.1 Prove: $\sim(P \vee Q) \equiv \sim P \wedge \sim Q$ (De Morgan's Law) (8 points)

P	Q	$P \vee Q$	$\sim(P \vee Q)$	$\sim P$	$\sim Q$	$\sim P \wedge \sim Q$
T	T					
T	F					
F	T					
F	F					

Are columns $\sim(P \vee Q)$ and $\sim P \wedge \sim Q$ identical? _____

J.2 Prove: $\sim(P \wedge Q) \equiv \sim P \vee \sim Q$ (De Morgan's Law) (7 points)

P	Q	$P \wedge Q$	$\sim(P \wedge Q)$	$\sim P$	$\sim Q$	$\sim P \vee \sim Q$
T	T					
T	F					
F	T					
F	F					

Are columns $\sim(P \wedge Q)$ and $\sim P \vee \sim Q$ identical? _____

ANSWER KEY - Section J

J.1 Answer:

P	Q	$P \vee Q$	$\sim(P \vee Q)$	$\sim P$	$\sim Q$	$\sim P \wedge \sim Q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Yes, columns $\sim(P \vee Q)$ and $\sim P \wedge \sim Q$ are identical, therefore the equivalence is proven.

J.2 Answer:

P	Q	$P \wedge Q$	$\sim(P \wedge Q)$	$\sim P$	$\sim Q$	$\sim P \vee \sim Q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Yes, columns $\sim(P \wedge Q)$ and $\sim P \vee \sim Q$ are identical, therefore the equivalence is proven.

EXAM PREPARATION WORKSHEET - SUMMARY

Total Points: 200

Section	Topic	Points
A	Identify Propositions	10
B	Complete Basic Truth Tables	30
C	Complex Truth Tables	25
D	Statement Analysis	15
E	Logical Classification	20
F	Set Fundamentals	20
G	Set Operations	30
H	Venn Diagram Problems	30
I	Multiple Choice	20
J	Prove Logical Equivalence	15
TOTAL		200

Study Tips:

- Review all answer keys and understand the reasoning behind each answer
- Practice drawing Venn diagrams for complex set problems
- Memorize the precedence of logical operators: $\sim$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$
- Remember De Morgan's Laws for set and logic operations
- Understand the difference between $\subseteq$ (subset) and $\subset$ (proper subset)
- Know when a proposition is a tautology, contradiction, or contingency

Good luck on your exam!

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